NICO EPLER

Master of Engineering (Electronic)

PROFESSIONAL SUMMARY

Electronic engineer (M.Eng Electronic *cum laude*, B.Eng Mechatronic) specialising in UAV systems and autonomous robotics. Currently UAS Engineer at Flying Robot (Pty) Ltd, leading system architecture, integration and flight validation for large-scale ISR quadcopter platforms. Core expertise spans robotic hardware and software development, autonomous systems development, ROS 2/SLAM-based autonomy, computer vision and sensor integration, complemented by embedded hardware design and PCB development. Published researcher in autonomous underground mapping, with an international track record spanning UAV systems engineering, mechatronic consulting and mechanical fabrication across diverse industrial and research settings.

CORE COMPETENCIES

- **Robotics, Autonomy and Perception** – Practical experience with ROS 2, visual-inertial odometry, SLAM, path planning, autonomous exploration frameworks, computer vision and image processing.
- **UAV Systems Engineering** – Extensive experience in UAV platform development, system architecture, subsystem integration, flight testing and operational validation.
- **System Integration and Validation** – Skilled in combining hardware, software, sensors and payloads into functional systems through simulation, testing, debugging and iterative improvement.
- **Embedded and Electronic Systems** – Experience in sensor interfacing, electronic circuit design, custom PCB design, communication interfaces and hardware validation.
- **Mechanical Design and Prototyping** – CAD modelling, design-for-manufacturing, additive manufacturing, fabrication drawings and practical prototyping with real-world assembly constraints.

PUBLICATIONS

- N. Epler and C. Fisher, "Autonomous Quadcopter for Rapid Underground Mine Mapping and Exploration," *MATEC Web of Conferences*, vol. 417, article 04019, 2025. doi: 10.1051/matec-conf/202541704019.

WORK EXPERIENCE

Unmanned Aerial Systems Engineer | Flying Robot (Pty) Ltd, South Africa
Oct 2025 – Present

- Designed UAV system architecture and carried out platform development, configuration, subsystem integration and validation, with a focus on large-scale ISR quadcopter platforms.
- Performed flight-control tuning, UAV test piloting and system analysis to improve platform reliability and operational performance.
- Designed electronic circuits and custom PCBs for embedded sensors.
- Worked extensively with communication interfaces, including serial, I^2C, CAN, networking and wireless communication links.

Graduate Research Engineer – Master's Thesis | Stellenbosch University, South Africa
2024 – 2025

- Designed and developed a UAV platform for autonomous underground mine mapping and exploration (GNSS-denied).
- Designed, configured and validated a complete UAV autonomy stack, combining embedded hardware, flight-control software, ROS 2 development, visual-inertial odometry and SLAM components.
- Developed an end-to-end autonomous exploration framework for underground environments.
- Tested and evaluated the system through simulation and real-world indoor flight experiments, focusing on mapping performance, autonomy and platform reliability.

UAV Systems and Product Consultant | Phantom Pilots, South Africa
Jan 2024 – Dec 2025

- Assisted with quadcopter component selection, system integration and testing for early-stage UAV product development.
- Consulted on and developed mechatronic side projects, providing practical design input, component-selection support, and system-level development.

Personal Details

+27 82 653 7180

nicoepler66@gmail.com

Zurich, Switzerland

Nico Epler | LinkedIn

nicoepler.com

Nationality:
Dual German and Namibian

Languages:
German – Native speaker
English – Fluent
Afrikaans – Fluent

Software and Programming

C, R and Python programming
PLC programming
MATLAB
ROS 2
Computer vision and image processing
Machine learning and reinforcement learning fundamentals
EasyEDA
Autodesk Inventor, Fusion and Onshape
Microsoft Office: PowerPoint, Word and Excel

Soft Skills

Strong work ethic
Honest and reliable
Organised, punctual and responsible
Team-oriented
Innovative
Eager to learn

Engineering Design and Drafting Intern | Metallum Fabrications, Namibia
2022 + 2023

- Held responsibility for the end-to-end development of steel assemblies, from CAD design and fabrication drawings through to cutting, welding, fitting and installation.
- Gained hands-on exposure to workshop practices, manufacturing drawings, practical fabrication constraints and tolerancing.

MENTIONABLE PROJECTS

- Designed and developed a quadcopter for autonomous mapping and exploration in GNSS-denied underground mine environments (M.Eng Electronic).
- Designed, constructed and evaluated a handheld 3D laser-line scanner (final-year B.Eng project – 81%).
- Designed and implemented a multi-functional light source using an STM microcontroller (Electronic Design).
- Designed and constructed an automatic coin sorter (Mechatronics; Group).
- Designed an internal thickener mechanism for an industrial slurry-settling tank, including drive-system sizing, torque analysis and motor specification (Machine Design; Group).
- Pump-system analysis and recommendation for a mine (Fluid Mechanics; Group).
- Design and part selection for a complete water pump, including manufacturing drawings (Machine Design).

EDUCATION

Stellenbosch University – M.Eng (Electronic), *cum laude* 2024–2025
Stellenbosch University – B.Eng (Mechatronic) 2020–2023
- Golden Key International Honour Society

Otjiwarongo Secondary School 2015–2019
- National top-8 academic performer in Namibia (Grade 12)
- School Representative Council Head Boy
- Dux Scholar throughout high school

Deutsche Privatschule Otjiwarongo 2008–2014

SPORT & CULTURAL ACHIEVEMENTS

Namibian National Inline Hockey Team 2016–2019
- Assistant Captain 2018, 2019

Interactors Board Member 2016–2018
- President 2018

Otjiwarongo Junior Town Council – Treasurer 2016–2017
Allan Gray Orbis Foundation Candidate Fellow (Selected fellowship recipient) 2020–2023
- 2nd place in the 2022 Jamboree Venture Pitch competition, securing grant funding for further development.

Six-Time Cape Town Cycle Tour Finisher – 109 km cycling event 2020–2026
Winner – SAICE Stellenbosch Bridge Building Competition 2022
Private Tutor – Applied Mathematics 2022
University Tutor – Electro-Techniques 2024–2025
University Teaching Assistant – Computer Programming 2024

INTERESTS & HOBBIES

Hiking, cycling, jogging, investing, travelling and exploring the outdoors.